

KARDZHALI, BulgariaWMO: **157300**Lat: **41.650N**Lon: **25.383E**Elev: **331**StdP: **97.41**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.5	-6.2	-12.5	1.3	-6.1	-10.1	1.7	-4.2	18.3	8.5	16.7	2.5	1.7	0	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.3	33.7	21.8	32.1	21.2	30.7	20.7	23.9	30.7	22.7	29.5	21.7	28.3	4.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	17.2	28.0	20.6	15.9	26.4	19.5	14.9	25.2	73.5	30.7	69.0	29.6	65.0	28.4	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	14.4	10.1	-12.3	36.8	2.5	2.4	-14.1	38.5	-15.6	39.9	-17.0	41.3	-18.8	43.0		
(5)			-12.7	24.7	2.3	1.8	-14.3	26.0	-15.7	27.0	-17.0	28.0	-18.7	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.1	2.4	4.2	7.7	12.2	16.9	21.0	23.8	24.0	19.3	13.6	8.4	3.6
	DBStd	8.45	4.62	4.79	4.14	3.60	3.04	2.93	2.54	2.51	3.29	3.90	4.26	4.76
	HDD10.0	817	239	168	94	19	1	0	0	0	0	13	78	204
	HDD18.3	2413	496	396	330	184	63	9	1	1	28	153	298	456
	CDD10.0	1961	2	5	23	86	214	330	427	432	279	124	31	7
	CDD18.3	514	0	0	0	1	18	89	169	174	57	5	0	0
	CDH23.3	4887	0	0	1	18	172	778	1663	1727	483	44	1	0
	CDH26.7	1862	0	0	0	2	24	247	694	744	144	7	0	0
Wind	WSAvg	2.9	2.4	3.3	3.8	3.5	3.0	2.6	3.2	3.0	2.6	2.6	2.6	2.7
Precipitation	PrecAvg	649	54	60	54	53	67	51	34	27	47	59	62	80
	PrecMax	1088	164	151	136	144	189	130	106	70	155	148	183	192
	PrecMin	387	2	3	6	9	21	5	0	0	3	6	1	0
	PrecStd	143	38	44	35	29	40	31	27	23	37	37	40	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	17.8	21.2	25.9	29.1	33.2	36.7	36.2	32.6	27.7	21.1	16.9
		MCWB	10.8	11.2	13.6	18.3	19.0	22.4	22.5	21.6	20.5	17.9	14.7	13.3
	2%	DB	12.9	15.4	18.6	22.6	26.9	31.0	33.8	33.9	30.3	24.1	18.6	14.8
		MCWB	9.4	10.3	12.4	15.2	18.1	21.5	22.0	21.7	19.4	16.6	13.8	12.2
	5%	DB	11.3	13.4	16.5	20.7	25.2	29.5	32.1	32.3	28.2	21.9	16.9	12.9
		MCWB	8.4	9.6	11.2	14.0	17.6	20.8	21.3	21.2	18.5	15.5	13.3	10.3
	10%	DB	9.4	11.6	14.5	18.7	23.6	27.8	30.4	30.8	26.1	19.9	15.1	10.7
		MCWB	6.9	8.6	10.1	12.9	16.8	20.0	20.6	20.6	17.7	14.8	12.3	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	12.6	14.6	18.9	21.0	24.4	25.4	25.3	22.9	18.8	16.0	14.2
		MCDB	13.8	15.5	19.0	24.9	26.7	30.8	32.6	32.2	31.0	24.5	18.9	15.9
	2%	WB	10.1	11.1	13.1	16.4	19.4	23.0	23.9	23.8	20.4	17.6	14.9	12.4
		MCDB	12.5	14.2	17.2	21.3	25.2	29.3	30.9	30.7	27.3	22.5	17.5	14.4
	5%	WB	8.6	9.9	11.9	14.7	18.3	21.8	22.6	22.6	19.3	16.5	13.7	10.6
		MCDB	10.8	13.2	15.5	19.4	23.8	27.5	29.4	29.6	25.9	20.5	16.3	12.6
	10%	WB	7.1	8.6	10.7	13.5	17.3	20.7	21.5	21.5	18.4	15.5	12.4	8.9
		MCDB	9.0	11.4	13.9	17.7	22.4	26.2	28.1	28.7	24.6	19.0	14.8	10.5
Mean Daily Temperature Range		MDBR	7.4	8.4	9.1	9.9	10.6	11.3	12.1	12.3	11.2	9.5	8.3	7.0
	5% DB	MCDBR	10.6	11.9	12.8	13.2	13.5	13.7	14.7	14.6	14.6	13.2	11.1	9.4
		MCWBR	7.1	7.4	7.4	7.0	6.4	5.8	5.7	5.6	6.2	6.6	6.7	6.4
	5% WB	MCDBR	9.1	11.0	11.4	12.0	12.2	12.4	13.2	13.0	13.0	11.7	9.5	7.9
		MCWBR	6.5	7.2	7.1	7.0	6.5	6.1	6.3	6.1	6.1	6.4	6.1	5.7
Clear-Sky Solar Irradiance	taub		0.340	0.373	0.422	0.478	0.459	0.461	0.459	0.466	0.443	0.402	0.368	0.346
	taud		2.343	2.240	2.125	1.981	2.108	2.147	2.164	2.136	2.166	2.262	2.311	2.317
	Ebn at Noon		793	817	816	795	823	821	817	797	786	775	749	746
	Edh at Noon		86	111	139	173	156	150	147	146	133	107	88	82
All-Sky Solar Radiation	RadAvg		1.68	2.47	3.54	4.69	5.83	6.36	6.70	6.11	4.50	2.98	1.92	1.48
	RadStd		0.22	0.28	0.32	0.52	0.40	0.54	0.47	0.33	0.32	0.30	0.20	0.22